C-Band 6×6 Dual-Polarised Antenna Array

Results summary and project status

Document	TM-RF-15-002, Revision A
Date	30 October 2026
Report type	Results and status report
Frequency band	4.0–7.0 GHz (C-band)
Prepared by	Telmart RF & Microwave Engineering
Status	In progress: full-array verification and element development

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1 Summary of results

5.35–6.26 GHz
Present element matched (VSWR ≤ 3): 16 % bandwidth

54 %
Bandwidth required for 4–7 GHz

8.7–9.8 dBi
Element gain (accepted power), 5–6.5 GHz

> 20 dB
Isolation between V and H inputs over most of the band (array)

Main result. The poor input match reported on the array is caused by the radiating element itself, whose matched bandwidth is about 16 %, well below the 54 % needed for 4–7 GHz. The radiator works correctly inside its band, which is why the gain is as expected while the VSWR is poor.

Consequence. A change to the feed network alone cannot bring the VSWR below 3:1 across 4–7 GHz. A new wideband radiating element is required; its development has started (section 5).

2 Design reviewed

Item	Value
Array	6 × 6 dual-linear-polarised stacked patches, 276.86 × 276.86 mm, pitch ≈ 44 mm
Inputs	Input 1: V-polarisation; Input 2: H-polarisation (one 36-way feed network each)
Stack-up	Ground / RT5880 0.8 mm / driven patches and feeds / 4 mm air / FR4 0.4 mm / parasitic patches
Reported symptom	Gain as expected; VSWR above 3:1 over most of 4–7 GHz

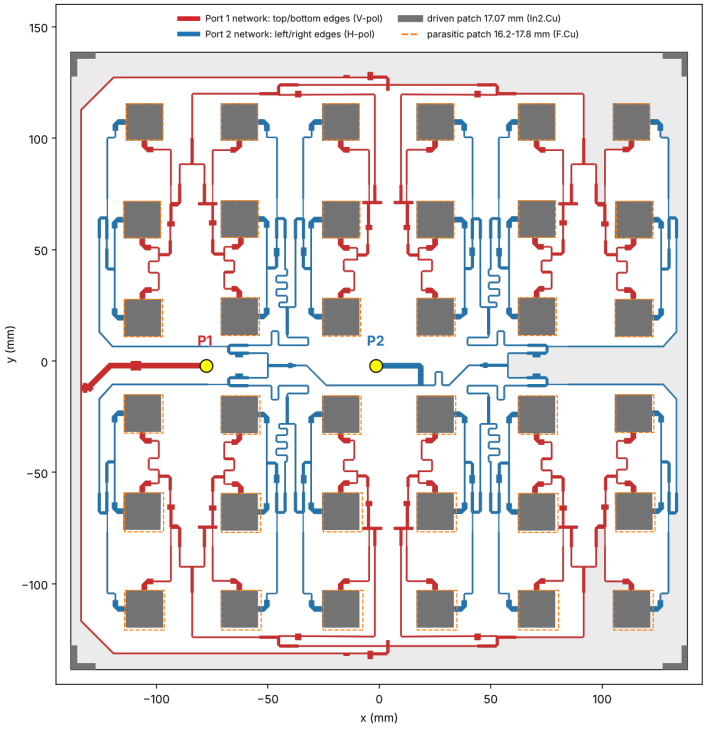


Figure 1. Array layout reviewed (client design). Red: input 1 feed network; blue: input 2 feed network.

3 Element results (present design)

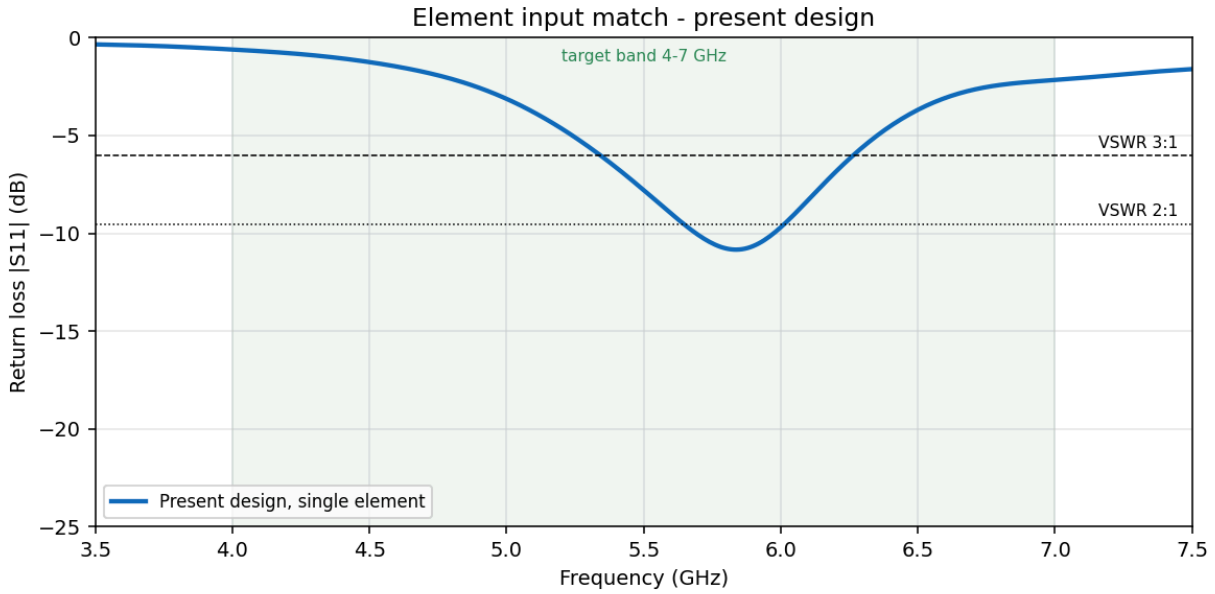


Figure 2. Input match of one element of the present design. Dashed line: VSWR 3:1; dotted: VSWR 2:1; shaded: 4–7 GHz target band.

Frequency (GHz)	Return loss |S11| (dB)	VSWR
4.0	-0.61	28.6
4.5	-1.25	13.9
5.0	-3.13	5.6
5.5	-7.79	2.4
6.0	-9.69	2.0
6.5	-3.69	4.8
7.0	-2.17	8.1

Frequency (GHz)	Gain, accepted power (dBi)	Realised gain incl. mismatch (dBi)
5.0	8.70	5.81
5.5	9.25	8.46
6.0	9.74	9.25
6.5	9.79	7.36

The element is matched only between 5.35 and 6.26 GHz. Outside this window the realised gain falls because of the mismatch, while the gain referred to accepted power stays at 8.7–9.8 dBi.

4 Complete array results (preliminary)

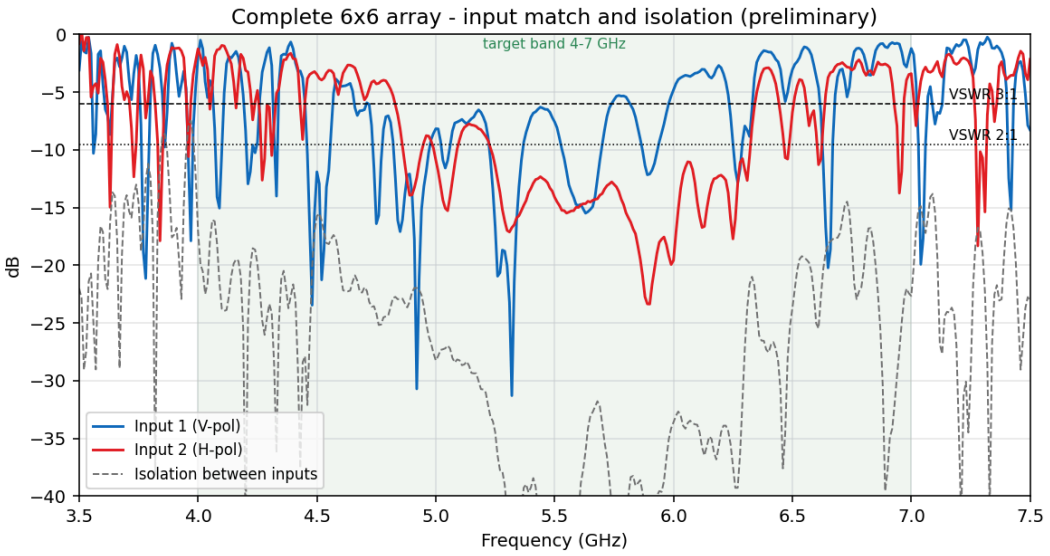


Figure 3. Complete 6×6 array: input match of both inputs and isolation between them. Preliminary: final values follow in the next revision.

- Both inputs are best in the middle of the band (about 5.0–6.2 GHz).
- Below about 4.8 GHz and above about 6.3 GHz both inputs are mostly at VSWR 3:1 or worse, consistent with the element result.
- Isolation between the V and H inputs is better than 20 dB over most of the band.
- The array result confirms that the element bandwidth is the limiting factor.

5 Element development (in progress)

A new wideband element is being developed to meet VSWR < 3 (target < 2) over the full 4–7 GHz band on both polarisations. Iteration 1 has been completed; it is matched from 4.95 to 5.75 GHz and has defined the design direction for iteration 2.

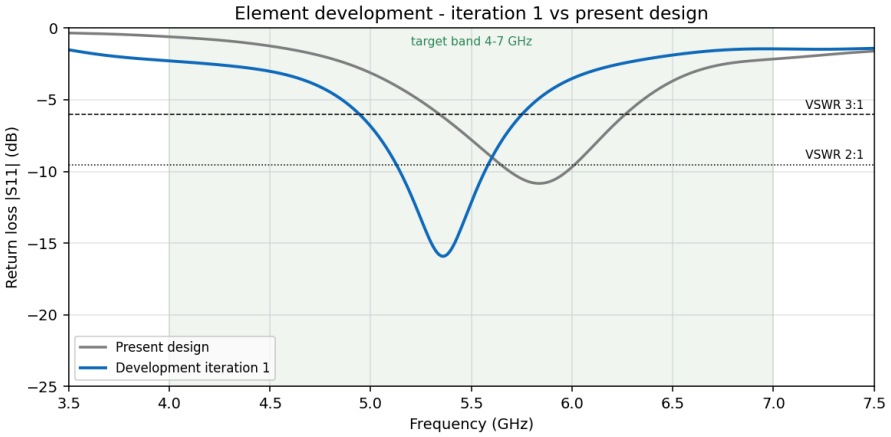


Figure 4. Element development: iteration 1 compared with the present design.

6 Project status

#	Work package	Status	Result / next output
1	Design data review (layout and stack-up)	● Completed	Design decoded, both feed networks traced
2	Element performance analysis, present design	● Completed	Root cause identified: element bandwidth ~16 %
3	Feed-network correction assessment	● Completed	Feed-network change alone is not sufficient
4	Complete-array verification, present design	◐ In progress	Preliminary results issued (section 4); final values in Rev. B
5	Wideband element development	◐ In progress	Iteration 1 completed; iteration 2 is the next step
6	Array integration of the new element	○ Planned	Updated array layout and simulated performance
7	Prototype and measurement	○ Planned	Measured input match, isolation and patterns

● completed ◐ in progress ○ planned

7 Actions requested from the client

- Confirm the required specification: VSWR limit (3:1 or 2:1), band edges, gain and isolation targets.
- Confirm any mechanical constraints for the new element (total height, materials, array size and pitch).
- If available, share measured S-parameters of the existing board for correlation.